



Gennoor Tech
TRAIN. INNOVATE. BUILD.

3-DAY INTENSIVE COURSE · 24 HOURS

RAG Engineering

Build retrieval-augmented generation systems for enterprise

3 24 15+ 5+

DAYS HOURS HANDS-ON LABS PROJECTS

www.gennoor.com · enquiry@gennoor.com

Course Overview

This 3-day intensive program covers everything from foundational concepts to advanced techniques, combining lectures, live demos, hands-on labs, and collaborative workshops for maximum learning impact.

Prerequisites: Python (intermediate), basic LLM API experience.

Who Is This For?



AI Engineers



Backend Developers



L&D Teams
Internal trainers



Technical Leads
Team leads and architects

Learning Outcomes

- 1 **Design and implement** — production RAG pipelines
- 2 **Select and optimize** — chunking, embedding, and indexing strategies
- 3 **Build with vector databases** — FAISS, Chroma, Pinecone, Azure AI Search
- 4 **Implement hybrid search**, — reranking, and citation tracking
- 5 **Evaluate RAG quality** — and implement guardrails

3-Day Curriculum

Interactive format with live demos, hands-on labs, and collaborative workshops. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

RAG Fundamentals & Chunking

HANDS-ON LABS + WORKSHOP

- SESSION A
- RAG Fundamentals
- RAG architecture: ingestion, indexing, retrieval, generation
 - Chunking: fixed, recursive, semantic, document-aware
 - Embedding models: OpenAI, Cohere, sentence-transformers
- SESSION B
- Chunking
- Lab: Compare chunking strategies on the same dataset
 - Document loaders for PDF, DOCX, HTML, and databases

02

Vector Databases & Hybrid Search

DEMOS + LABS + DISCUSSION

- SESSION A
- Vector Databases
- FAISS, Chroma, Pinecone, Weaviate — architecture comparison
 - Azure AI Search: indexes, skillsets, vectorization
 - Hybrid search: keyword + semantic + reranking
- SESSION B
- Hybrid Search
- Metadata filtering and multi-tenant patterns
 - Lab: Hybrid search RAG with Azure AI Search

03

Production RAG & Evaluation

LECTURE + HANDS-ON LABS

- SESSION A
- Production RAG
- Citation tracking and source attribution
 - Query routing and multi-index strategies
 - RAG evaluation: faithfulness, relevancy, completeness
- SESSION B
- Evaluation
- Guardrails: hallucination detection, content safety
 - Lab: Production RAG system with evaluation pipeline

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools demonstrated and practiced hands-on during the course

- Azure AI Search
- Azure AI Services
- ChromaDB
- FAISS

What You Take Home



RAG Pipeline Templates
Included with course completion



Vector DB Comparison Matrix
Included with course completion



Evaluation Framework
Included with course completion



Chunking Strategy Guide
Included with course completion

Certification of Completion



Gennoor Tech — Certificate of Completion
Upon completing all 3 days (24 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "RAG Engineering". Digitally signed by the Founder & Director. Validates proficiency in all course topics and hands-on competency.

Recommended Next Certifications

AI-103
Azure AI Solutions

DP-100
Azure Data Scientist

AI-901
Azure AI Fundamentals (New 2026)

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited

Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of **16+ active Microsoft certifications** experience. Delivered AI and technology programs for Microsoft, EY, IBM, Boeing, including AB-100 (Agentic AI), AI-103, AZ-400, Saudi Aramco, HDFC Bank, and 80+ organizations across 6+ countries including the AI-300, PL-400, and GitHub certifications. 10-day AI Leadership program for MCIT Saudi Arabia.

Builder of open-source MCP servers and Copilot Studio integrations. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options

 **On-Site** — customized at your office |  **Virtual Live** — interactive via Teams/Zoom |  **International** — GCC, Africa, APAC, UK



Ready to Start Your RAG Engineering Journey?

Contact us to discuss a customized program
for your organization or register for the next open batch.

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